

Problem – Solution Fit Template

Date	13 July 2026	Maximum Marks	2 Marks
Project Name	India's Agricultural Crop Production Analysis using Tableau		

Problem – Solution Fit: The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps identify behavioral patterns and recognize what would work and why.

1. Problem – Solution Fit Canvas

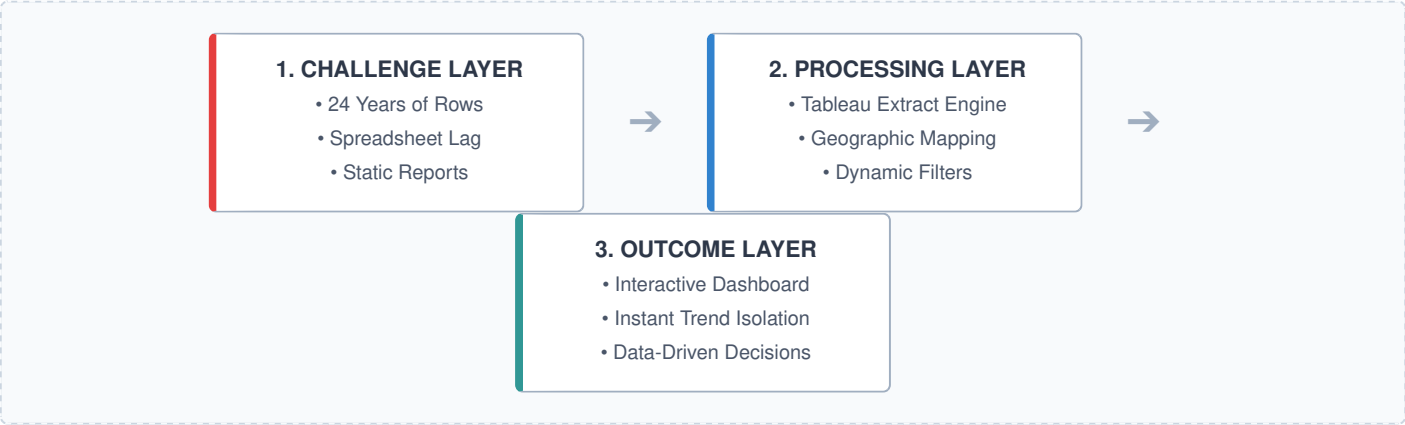
1. CUSTOMER SEGMENT(S) [CS] Who is your customer? Agricultural analysts, researchers, state-level farming executives, and policymakers at SmartBridge / agricultural ministries managing multi-decade crop production metrics.	6. CUSTOMER CONSTRAINTS [CC] What constraints prevent action? No software programming expertise (cannot use R/Python for pipelines); restricted to zero-budget frameworks (Tableau Public); no server/backend modifications allowed.	5. AVAILABLE SOLUTIONS [AS] Which solutions are available? Manual Excel files, static summary PDFs, traditional tabular databases. Pros: highly customizable rows. Cons: fails on big files, no interactive cross-filtering, no spatial maps.
2. JOBS-TO-DONE / PROBLEMS [J&P] Which problems do you address? • Huge agricultural dataset hard to analyze • Manual spreadsheet calculations • No interactive visualizations • Difficult crop/state comparisons • Lack of historical trend analysis	9. PROBLEM ROOT CAUSE [RC] What is the real reason it exists? The Indian Crop Dataset contains thousands of logs across 1997–2021. Plain row grids limit data scanning, making multi-axis seasonal evaluations highly complex without dynamic visual computing engines.	7. BEHAVIOUR [BE] What does your customer do to solve it? Writing hundreds of complex Excel VLOOKUP formulas, manually generating static charts monthly, and cross-checking multiple disconnected sheets to calculate regional yield aggregates.
3. TRIGGERS [TR] / 4. EMOTIONS [EM] What triggers action? Before/After? Triggers: Sudden system crashes during massive file load. Before: Stressed, slow output. After: Empowered, confident, immediate data access.	10. YOUR SOLUTION [SL] How does your solution fit reality? A dedicated Tableau Public BI Dashboard utilizing KPI cards, interactive dynamic filters, spatial state maps, seasonal matrix trees, and a comprehensive chronological Tableau Story layout.	8. CHANNELS OF BEHAVIOUR [CH] Online / Offline channels used? Online: Navigating intranet shared drives to fetch csv rows, emailing attachments. Offline: Manual team synchronization to sort conflicting crop totals.

2. Problem – Solution Analysis Matrix

Problem	Existing Challenges	Proposed Tableau Solution	Benefits	Expected Outcome
Large dataset difficult to analyse	Processing 1997–2021 logs freezes traditional text software.	High-performance Tableau data extraction and structural optimization.	Eliminates loading lag; executes rendering instantly.	Immediate analytics access across all historical crop records.

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Manual spreadsheet analysis	High risk of corrupted formulas, accidental changes, and high processing times.	Pre-built calculated fields and automated visual distribution models.	Eliminates human error; ensures rapid delivery metrics.	Zero dependence on manual file management or manual styling.
No interactive visualizations	Static reporting hides outliers and restricts real-time interrogation.	Dynamic interactive filtering and cross-dashboard action links.	Enables users to filter variables freely without rebuilding assets.	Self-service analytical framework for non-technical leadership.
Difficult comparison between states and crops	Cross-tabulating regional metrics requires large pivot aggregations.	Geospatial State Maps combined with Side-by-Side Bar Charts.	Instant geographic identification of high and low yield centers.	Flawless crop yield and surface acreage benchmarking.
Lack of trend analysis	Hard to track cyclical shifts over 24 years across changing weather seasons.	Chronological Line Visuals and Dedicated Seasonal Dimensions.	Highlights long-term patterns, historical booms, and structural drops.	Proactive policy alignment with historic cultivation cycles.
Difficult decision making	Decisions delayed due to unstructured tables and complex datasets.	High-level KPI Blocks unified within a structured Tableau Story.	Presents a clear dashboard truth for strategic interventions.	Rapid, data-validated executive agricultural policymaking.

3. Visual Problem-to-Solution Flow



4. Conclusion

The deployment of the Tableau Public framework successfully bridges the gap between massive historical data records and comprehensive strategic planning. By migrating the 1997–2021 Indian Agricultural Crop Production Dataset away from manual, static spreadsheet architectures, this solution establishes a responsive data delivery environment. Non-technical stakeholders can freely drill down by region, evaluate seasonal yield differentials, isolate micro-trends, and finalize immediate, data-validated agricultural decisions without any underlying software engineering barrier.